

# BioSpecInfo

## BioSpecInfo — Technical Dossier

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Versione **bsi-v187** · commit `b975758`

Generato il 2026-10-07

Documento generato automaticamente dal sorgente Markdown versionato nel repository. La versione e il commit stampati in ogni piè di pagina identificano esattamente lo stato del software descritto.

# BioSpecInfo — Technical Dossier

Open-access web platform for computational chemistry and the interactive consultation of chemical data on any device.

|                |                                                                                                                             |
|----------------|-----------------------------------------------------------------------------------------------------------------------------|
| Author         | Samuele Pio Provenzano                                                                                                      |
| Thesis advisor | Prof. Savino Longo — University of Bari Aldo Moro                                                                           |
| Profile        | Cheminformatics Engineer / Scientific Software Developer                                                                    |
| Software type  | Progressive Web App (PWA), client-side, local-first, offline-first                                                          |
| Live demo      | <a href="https://samupropio1-ship-it.github.io/BioSpecInfo-v11/">https://samupropio1-ship-it.github.io/BioSpecInfo-v11/</a> |
| Repository     | <a href="https://github.com/samupropio1-ship-it/BioSpecInfo-v11">github.com/samupropio1-ship-it/BioSpecInfo-v11</a>         |

This dossier summarises the application and is the cover of the full technical documentation (SAD, Validation & Benchmark, Security & Compliance, Licensing) attached below.

## 1. Executive summary

BioSpecInfo is an **interactive cheminformatics platform** that runs all of its scientific logic **in the client browser**, with no server: molecular analysis, spectral prediction, 2D/3D modelling and educational tools — all **free, installable as an app and working offline**.

Its distinctive value is the **rare combination of advanced chemistry and modern software engineering** (WebAssembly, WebGL, PWA), with spectral predictors **quantitatively validated** against the official SDBS and NIST databases.

## 2. Product factsheet

**Value proposition** — A **≈4.5 MB** web platform (main file) that runs molecular analysis, spectral prediction and 3D visualisation at **zero server cost** and **100% offline-ready**.

| Technical architecture                                              | Content & capabilities                                                                               |
|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Client-side / Local-First (no backend)                              | <b>RDKit</b> cheminformatics engine (WebAssembly)                                                    |
| Single-file HTML5 / ES6, installable as <b>PWA</b>                  | <b>Spectra Centre</b> : IR, <sup>1</sup> H/ <sup>13</sup> C-NMR, MS + isotope pattern, UV-Vis, Raman |
| <b>2D</b> (SVG/Canvas) and <b>3D</b> (three.js, 3Dmol.js) rendering | <b>Organic Chemistry</b> : 25 interactive modules                                                    |
| <b>Service Worker</b> offline-first, versioned                      | <b>Retrosynthesis</b> : 46 strategies + animated 2D/3D mechanisms                                    |

| Technical architecture                          | Content & capabilities                                            |
|-------------------------------------------------|-------------------------------------------------------------------|
| Zero copleft dependencies (MIT/BSD-3 only)      | <b>3D Drug Atlas:</b> 63 diseases (23 tumours), 3D organs, PubMed |
| Local storage (localStorage, SQLite via sql.js) | 3D astrochemistry, Quantum Lab, Academy (SM-2 quizzes)            |

### Quantitative validation (thesis benchmark)

- **<sup>1</sup>H-NMR** vs SDBS → **MAE 0.31 ppm** (89% of signals within  $\pm 0.5$  ppm)
- **IR (C=O band)** vs NIST WebBook → **MAE 11 cm<sup>-1</sup>** (90% within  $\pm 30$  cm<sup>-1</sup>)
- Interactive performance **30-60 fps** from iPhone/iPad to Raspberry Pi 4

## 3. Verified capabilities (current code state)

**Spectra Centre (molecular editor & analysis)** — draw from scratch (full periodic table, templates, drag-to-bond) or load via SMILES/library →

- All chemical notations: formula, canonical SMILES, CXSMILES, SMARTS, InChI, InChIKey, mol block, common/IUPAC name, CAS (one-click copy).
- Predicted spectra with bibliographic sources; **isotope pattern** (IUPAC abundances) and MS **adduct masses**.
- Functional groups and properties (H donors/acceptors, aromatic, rotatable, stereocentres) **highlightable on the 2D and 3D structure**.
- Drug-likeness: **Lipinski, Veber, Egan, Ghose, Muegge**, ESOL solubility, **QED, Brenk/PAINS structural alerts**, physicochemical radar.

- **Tanimoto** similarity (Morgan ECFP4 fingerprint), elemental analysis, SMARTS substructure search, SVG/PNG/MOL export.

**Synthesis & Retrosynthesis** — 46 strategies with an **animated mechanism player** in **2D and 3D**: *electron-pushing* arrows, reaction intermediates, and a retrosynthesis mode showing **synthon formation**.

**Organic Chemistry** — 25 modules (fundamentals → heterocycles, pericyclic & Woodward-Hoffmann, organometallics & cross-coupling, radicals, spectroscopy, retrosynthesis & protecting groups).

**Data Science** — 3D Drug Atlas (13 body regions, 63 diseases of which 23 tumours, 3D organ models, PubMed links), ML labs, SQL playground.

**Other** — 3D astrochemistry, Quantum Lab, Academy with spaced repetition (SM-2), interactive simulations, databases of elements/drugs/biomolecules.

Overall metrics declared in the thesis (historical data assets): 324 organic reactions in vector SVG graphics, 26 biosynthetic pathways, 140+ drugs, 80+ biomolecules, 118 elements, 10 quantum simulators.

## 4. Technology stack

HTML5 · ES6/ESNext · WebAssembly (RDKit MinimalLib, sql.js/SQLite) · WebGL (three.js, 3Dmol.js) · SVG / Canvas 2D · Service Worker / PWA · public API integration (PubChem PUG-REST, NIST, SDBS, MassBank, nmrshiftdb2, NASA/ESA).

## 5. Author's declaration (self-attestation)

The undersigned **Samuele Pio Provenzano** declares that:

1. The software **BioSpecInfo** is the author's **original work**, except for the third-party open-source libraries (RDKit, 3Dmol.js, three.js, SmilesDrawer, sql.js) used under their respective permissive licenses and documented in the *Open-Source Licensing & IP Matrix*.
1. The reported validation data (MAE 0.31 ppm for  $^1\text{H}$ -NMR vs SDBS; MAE 11  $\text{cm}^{-1}$  for the IR C=O band vs NIST) were measured by the author as part of the thesis work; the predictors are **heuristic group-additivity** models, suitable for educational use and qualitative functional-group identification, with the limitations stated in the *Validation Report*.
1. The architecture is **local-first**: no user personal data is transmitted to author-controlled servers (alignment with GDPR principles).

This is an **author's declaration**, not a certification issued by a third party.

## 6. Attached documentation

1. **Software Architecture Document (SAD)** — architecture, components, technologies.
  2. **Verification & Validation & Benchmark Report** — test method, benchmarks, limitations.
  3. **Security, Privacy & Compliance** — local-first, GDPR, OWASP, GAMP 5 (alignment).
  4. **Open-Source Licensing & IP Matrix** — dependencies, licenses, intellectual property.
- \_Software license: proprietary — "All rights reserved" (see [LICENSE](#)).\_